

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. Engineering in CSE

Midterm Examination, Autumn 2025

Course Code: ACC 2401

Course Title: **Financial Managerial Accounting**

Time: 1 hour 30 minutes

Full Marks: 30

Answer all the questions. The figures in the right-hand margin indicate full marks.

Course Outcomes (COs), Program Outcomes (POs) and Bloom's Levels (BL) of the Questions			
CO	CO Statements	PO	BL
CO1	Explain the basic concept of financial accounting, cost accounting and management accounting.	PO11	C2
CO2	Analyze the basic concept of Cost Accounting and preparation of Cost Sheet.	PO11	C4
CO3	Apply the tools from accounting and cost accounting this would facilitate the decision making i.e. Budgeting, Make or Buy decision.	PO11	C3
CO4	Compare the different business situations and suggest to best solution with analytical abilities.	PO11	C5

Bloom's Levels (BL) of the Questions						
Letter Symbols	C1	C2	C3	C4	C5	C6
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1)	a)	How would you define the term "Accounting" and who are the users of it? Describe the accounting equation briefly.	CO1	C2	5
1)	b)	Is the capital introduced by the owner considered a liability of the business? , and how would you define the term internal liability and external liability in this context?	CO1	C2	5
2)	a)	Mr. Azim owns and operates an interior design studio called Azim Interiors. The following amounts summarize the financial position of her business on December 31, 2022: During September 2022 the following events occurred: a. Mr. Azim inherited \$15,000 and deposited the cash in the personal bank account. b. Performed services for a client . c. Paid off the beginning balance of accounts payable valued \$1,0000. d. Purchased supplies on account, \$500. e. Collected cash from a customer on account, \$1,000. f. Invested personal cash of \$1,000 in the business. g. Consulted on the interior design of a major office building and billed the client for services rendered, \$2,400. h. Paid office rent - \$900 and Paid advertising - \$100. i. Sold supplies to another business for \$150 cash, this was the cost of the supplies. j. Withdrew cash of \$(Last Four digit of your student ID) for personal use. k. Collected cash from his friend for buying a personal car \$1,00,000. l. Reed deposited \$15,000 cash to his personal bank account by selling some common stock. Required: 1) Record each transaction in the journal. 2) Post the transaction to the ledger with label the balance of each account.	CO2	C4	10

3)	a)	Melvin Dexter recorded the following events during the organizing phase of his new business and its first month of operations. May-4: Dexter received \$50,000 cash from his former partners in the law firm from which he resigned. May-5: Dexter deposited \$50,000 cash in a new business bank account entitled Melvin Dexter, Attorney. May-6: Dexter paid \$300 cash for letterhead stationary for his new law office. May-7: Dexter purchased office furniture for his law office. He agreed to pay the account payable, \$5,000, within six months. May-10: Dexter sold 500 shares of IBM Stock, which he and his wife had owned for several years, receiving \$75,000 cash from his stock broker. May-11: Dexter deposited \$75,000 cash from sale of the IBM stock in his personal bank account. May-12: A representative of a large company telephoned Dexter and told him of the company's intention to transfer its legal business to the new entity of Melvin Dexter, Attorney. May-29: Dexter finished court hearings on behalf of a client and submitted his bill for legal services, \$4,000. Dexter expected to collect from this client within two weeks. May-30: Dexter paid office rent expense, \$1,000. May-31: Dexter withdrew \$2,000 cash from the business for personal living expenses. Required 1. Analyze the effects of the above transactions on the accounting equation of Melvin Dexter, Attorney. 2. Prepare the income statement and statement of owner's equity of Melvin Dexter, Attorney based on the above transaction.	CO2	C5	10
OR					
3)	a)	Andrea Scarlett is a realtor. She organized her business as a corporation, Andrea Scarlett, Realtor, P.C. (Professional Corporation) 1. The business owes \$61,000 on a note payable for land that the business acquired for a total price of \$83,000. 2. The business spent \$23,000 for a Zinka Banker real estate franchise, which entitles the business to represent itself as a Zinka Banker office. This franchise is a business asset. 3. Scarlett owes \$80,000 on a personal mortgage for her personal residence, which she acquired in 2012 for a total price of \$160,000. 4. Scarlett has \$5,000 in her personal bank account, and the business has \$9,000 in its bank account. 5. Scarlett owes \$(Last Four digit of your student ID) on a personal charge account with Chico's. 6. The office acquired business furniture for \$15,000 on Sep 25. Of this amount, the business owes \$2,000 on account at September 30. 7. Office supplies on hand at the real estate office total \$1,300. 8. Received partial payment from client on account, \$2700 9. Received \$1,500 cash for helping a client. 10. Scarlett received a birthday gift from his friend valued \$3000. Required I. Analyze the effects of the above events on the accounting equation. II. Prepare the income statement, statement of owner's equity and balance sheet based on the above transaction.	CO2	C5	10

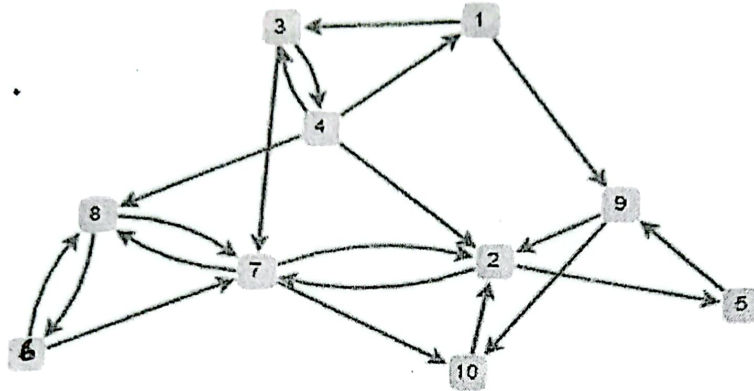
International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Mid Term Examination, Autumn-2025
Course Code: MATH-2407, Course Title: Mathematics-IV

Time: 1:30 Hours

Marks: 30

[Answer the following questions. Figures in the right margin indicates full marks]

- | | | | |
|--|------------|-------------|----------|
| 1. a) Let the function $f: \mathbb{R}^{\#} \rightarrow \mathbb{R}^{\#}$ be defined by $y = f(x) = x^2 + x - 4$ then find the value of $f^{-1}(10)$ | Marks
2 | CLO
CLO1 | DL
C2 |
| b) | 3 | CLO1 | C2 |



Determine whether the above relation is Anti-symmetric and Transitive or not

- | | | | |
|--|---|------|----|
| c) Prove $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ where n is a positive integer | 5 | CLO3 | C2 |
| Or | | | |
| If $(1 + i\frac{x}{a})(1 + i\frac{x}{b})(1 + i\frac{x}{c})\dots\dots\dots = A + iB$, Then prove that | 5 | CLO3 | C2 |
| $(1 + \frac{x^2}{a^2})(1 + \frac{x^2}{b^2})(1 + \frac{x^2}{c^2})\dots\dots\dots = A^2 + B^2$ | | | |

- | | | | |
|---|---|------|----|
| 2. a) Map the straight line joining $A(-2 + j)$ and $B(3 + j6)$ in the z -plane on to the w -plane when $w = u + jv = f(z) = 3 + j2z$ | 6 | CLO1 | C2 |
| b) Test the function $u(x,y) = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic function or not. | 2 | CLO1 | C2 |
| c) Find the poles and residue of the function $f(z) = \frac{z}{z^2 + 1}$ | 2 | CLO2 | C2 |

- | | | | |
|---|---|------|----|
| 3. a) Evaluate $\int_C \frac{1}{z} dz$ where C is the circle $x = \cos t, y = \sin t$ with $0 \leq t \leq 2\pi$ | 5 | CLO3 | C3 |
| b) Show that $\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i$, where c is the circle $ z = 5$ | 5 | CLO3 | C3 |

Or

- | | | | |
|---|---|------|----|
| Show that $\oint_c \frac{e^{tz}}{z^2 + 1} dz = 2\pi i \sin t$, where c is the circle $ z = 3$ | 5 | CLO3 | C3 |
|---|---|------|----|

International Islamic University Chittagong
Department of Computer Science and Engineering

B. Sc. in CSE Mid-term Examination, Autumn 2025

Course Code: **CSE 2423**

Course Title: **Database Management System**

Total marks: **30**

Time: **1 hour 30 minutes**

			CLO	DL	
Q1.	a)	Compare the three-level data abstraction.	2	1	2
	b)	Describe the major functions of the Query Processor in a DBMS. Explain how it interacts with the Storage Manager during query execution.	5	2	2
	OR	State the functions of a DBA. Differentiate between two-tier and three-tier database architectures with their applications.			
	c)	Differentiate between centralized, distributed, and NoSQL database systems. Mention one real-world application for each.	3	1	2
Q2.	a)	An event management company organizes various programs such as conferences, weddings, and concerts. For each event, details like date, venue, client, and assigned staff are recorded. Vendors provide services such as catering, sound, or decoration for the events. Staffs are assigned specific roles for particular events. Design an ER diagram that represents how the system stores and connects all this information.	5	3	3
	OR	An educational website allows users to register, enroll in different courses, and track their progress. Each course has several lessons and may be created by one or more instructors. Students can post reviews and ratings after completing a course. Design an ER diagram that models this system. Your design should clearly represent all main entities, their key attributes, and the relationships among them.			
	b)	Show with the diagram, how <i>aggregation</i> can be applied when modeling the relationship between Doctor, Patient, Appointment, and Employees entities	2	2	4
	c)	Explain with how different types of attributes represented in an ERD can be implemented in an actual database schema.	3	2	2
Q3.	a)	Write the reason for converting SQL queries into relational algebra during query processing	1+9	2	3

Consider the following schema:

Employee(**EID**, EName, DeptID, Salary)

Department(**DeptID**, DName, Location)

Project(**PID**, PName, DeptID)

WorksOn(EID, PID, Hours)

Write **relational algebra expressions** for:

- i. List the names of employees working on the project named "AI Research".
- ii. Find employees who do not work on any project.
- iii. Find the employees who works on the project of 'CSE department'

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 2025

Course Code: CSE-241

Course Title: Computer Algorithms

Total marks: 30

Time: 1 hours 30 minutes

[Figures in the right-hand margin indicate full marks]

Course Outcomes and Bloom's taxonomy domain and levels are indicated in additional columns]

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|-----|---|-----|----|---|
| 1a) | Find an asymptotic bound for the following using master method.
$T(n) = 5T(n/3) + n^2$
OR
Find an asymptotic bound for the following using recursion tree method.
$T(n) = 2T(n/3) + n^2$ | C01 | C2 | 3 |
| 1b) | Assume that $f(n) = 4n + 20$ and $g(n) = n * \log_{10} n$. Is $f(n)$ a lower bound of $g(n)$?
Prove or disprove this. | C02 | C6 | 3 |
| 1c) | Find how many times each line executes. Express the running time of each function in asymptotic notation.
<div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p>F()</p> <pre> for i = 1 to n j = n while j >= 1 j = j/2 </pre> </div> <div style="width: 45%;"> <p>G()</p> <pre> for i = 1 to $\log_2 n$ j = 1 to i print(i+j) </pre> </div> </div> | C01 | C6 | 4 |
| 2a) | Remove the largest element from the following heap and show step by step procedure: {20, 15, 7, 6, 9, 2, 6, 5, 1, 7, 8}.
OR
Insert a new element X+Y [X and Y are last two digits of your ID] into the following maxheap and show step by step procedure: {9, 8, 7, 6, 7, 2, 6, 5, 1}. | C02 | C3 | 3 |
| 2b) | Show the steps of the first PARTITION operation of the quick sort algorithm on the array A = {8, 2, 6, 1, 7, 3, 4}. Assume the last element (i.e. 4) is the pivot. | C01 | C5 | 3 |
| 2c) | What will be the recurrence for expressing the time taken by quicksort in the following two cases.
i) Input is sorted in decreasing order
ii) Each time the partition splits the input in 7:2 ratio
What will be the asymptotic bound for these cases? | C02 | C4 | 4 |
| 3a) | Prove that the problem of finding the longest common subsequence has optimal substructure property.
OR
Show that the matrix chain multiplication problem has overlapping subproblems property. | C01 | C3 | 3 |
| 3b) | Which of the following functions can be evaluated by dynamic programming? Here, X = last digit of your ID, $Y = \lfloor (X/3) \rfloor + 1$. Justify your answer.
$f(n) = f(n-Y) + n^2$
$g(n) = g(n/2) + g(n/4) + Y$
You may assume any suitable base cases for these functions. | C02 | C4 | 3 |
| 3c) | Given a sequence $p = \langle 3, 2, 4, 5, 6 \rangle$ which represents the chain of matrices such that i-th matrix A_i is of dimension $p_{i-1} \times p_i$. In response to a question asking to find the optimal parenthesization of multiplying this matrix chain, a student's answer was $(A_1 A_2)(A_3 A_4)$. Is his answer correct? Justify your answer by showing necessary calculations.
OR
Find the longest common subsequence of the following strings: X = "ATGCAGC" and Y = "TACGTC". From this result, find the LCS length of the following pairs of strings?
i) "ATGCAG" and "TACGT"
ii) "ATGC" and "TACGT" | C01 | C4 | 4 |

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 2025

Course Code: CSE-2427

Course Title: Theory of Computation

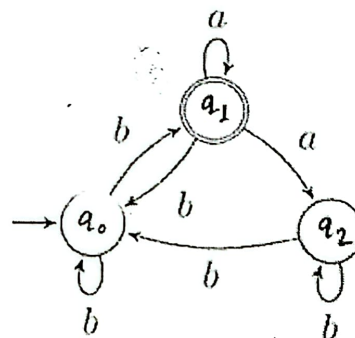
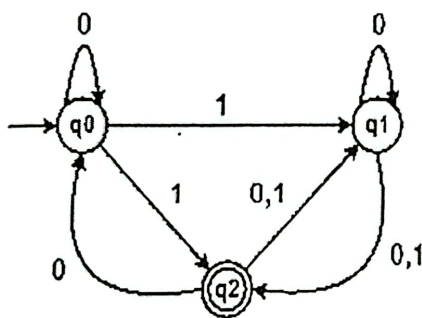
Total marks: 30

Time: 1 hours 30 minutes

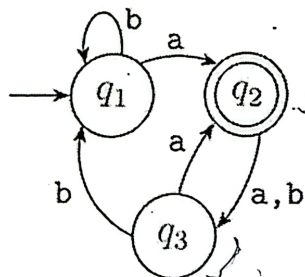
[Figures in the right-hand margin indicate full marks]

Course Outcomes and Bloom's taxonomy domain and levels are indicated in additional columns]

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|-----|--|-----|----|---|
| 1a) | Define Computability Theory and Automata Theory.
OR
Define Finite Automata. What are the components of a DFA? | C01 | C2 | 2 |
| 1b) | Construct DFA for the following languages where alphabet is $\{0, 1\}$.
i. $\{w \mid w \text{ begins with '01' and the length of } w \text{ is even}\}$
ii. $\{w \mid w \text{ contains '01' and has odd number of 1's}\}$ | C02 | C6 | 4 |
| 1c) | Write regular expressions for the languages described in 1(b). | C01 | C6 | 4 |
| 2a) | Design an NFA over alphabet $\Sigma = \{a, b\}$ such that it accepts every string starting and ending with the same symbol. Write an equivalent regular expression.
OR
Write one sentence description for each of the following languages.
i. $((0 1)(0 1)(0 1)(0 1))^*$
ii. $(10 \cup 0)^*$ | C02 | C3 | 2 |
| 2b) | Prove that the class of regular languages is closed under union.
OR
Prove that the class of regular languages is closed under concatenation. | C01 | C5 | 4 |
| 2c) | Convert any one of the following NFA to an equivalent DFA. | C02 | C4 | 4 |



- | | | | | |
|-----|--|-----|----|---|
| 3a) | Convert the following regular expressions to an NFA: $[a \cup ba (a \cup b)]^*(ba)^*$
OR
Give the transition table for the machine in question 3b. What sequence of states does it go through on input babb and abba? Does the machine accept the strings? | C01 | C3 | 2 |
| 3b) | Convert the following DFA into an equivalent regular expression. | C02 | C4 | 4 |



- | | | | | |
|-----|---|-----|----|---|
| 3c) | State the pumping lemma for regular expression. Use it to determine if the following language is regular. | C01 | C4 | 4 |
|-----|---|-----|----|---|

$A = \{w \mid w \text{ has } m \text{ 0s followed by } 2m \text{ 1s}\}.$

International Islamic University Chittagong
Center for General Education (CGED)
Midterm Examination Autumn- 2025
Course Code: GEBL-2401
Course Title: Bangla Language and Literature
Full Marks: 30 Time: 1.5 Hours

ক-বিভাগ

ভাষা ও নির্মিতি : ২০

(ডান পাশের সংখ্যা প্রশ্নের মান জ্ঞাপক)

প্রশ্ন নং	বর্ণনা	মান	CLO	Cognitive learning
০১.	“ইন্দো- ইউরোপীয় ভাষা গোষ্ঠী থেকে প্রাকৃত ও অপভ্রংশের মধ্য দিয়ে বাংলা ভাষার জন্ম” উক্তির আলোকে বাংলা ভাষার উৎপত্তি ও বিকাশের ইতিহাস আলোচনা কর।	১০	CLO1	Create
০২.	নিচের পরিভাষাগুলোর পরিচয় দাও: স্বরধ্বনি, বাংলা মৌলিক স্বরধ্বনি, স্পর্শ-ধ্বনি, তরল ধ্বনি, ঘোষ ধ্বনি।	১০	CLO1	Underst
অথবা	“আগামীর নেতৃত্ব ও বাংলাদেশ বিনির্মাণে ছাত্রসংসদের ভূমিকা”- শীর্ষক একটি বক্তব্য রচনা কর।	১০	CLO1	Crea

খ-বিভাগ

সাহিত্য : ১০

প্রশ্ন নং	বর্ণনা	মান	CLO	Cognitive learning
০৩.	“জীবনে এমন কত বিচ্ছেদ, কত মৃত্যু আছে ফিরিয়া ফল কী, পৃথিবীতে কে কাহার।”-উক্তির আলোকে পোস্টমাস্টার গল্পটি বিশ্লেষণ কর।	১০	CLO2 CLO3	Evaluat